Name ____ Date ___________ Period _____

Transformations of $f(x) = |x|$, Part II

TEACHER KEY | Algebra II Honors | Unit 2 | Day 3 | TEK A2.6.C

Objective: I can derive why (c, d) is always the vertex of $f(x) = a|x-c|+d$, and work backward from a vertex and a point to find the equation.

Vocabulary

Horizontal shift — $f(x-c)$; shifts opposite the sign of c.

Vertical shift — $f(x)+d$; shifts the same direction as the sign of d.

Vertex form — $f(x) = a|x - c| + d$, written specifically to make the vertex **(c, d)** easy to read off.

Part 1 — Prove the Vertex Is Always (c, d)

Evaluate $f(x) = a|x - c| + d$ at $x = c$, for any a, c, d:

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$f(c) = a|c - c| + d = a|0| + d = a \cdot \mathbf{0} + d = \mathbf{d}$ → so (c, d) is always ON the graph

Explain why (c, d) is the LOWEST point when $a>0$ (or highest when $a<0$), not just A point on the graph.

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Sample: **since $|x - c| \geq 0$ always, $a|x - c| \geq 0$ when $a>0$, so $f(x) \geq 0 + d = d$ for every x - meaning d is the smallest output**

Part 2 — Working Backward: Vertex + One Point → Equation

A function has vertex (3, -2) and passes through (5, 4). Find a, then write the equation.

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$a|5 - 3| + (-2) = 4$ → $a(2) - 2 = 4$ → $2a = \mathbf{6}$ → $a = \mathbf{3}$

Equation: $f(x) = \mathbf{3|x - 3| - 2}$

You Try. Vertex (-1, 5), passes through (2, -4). Find a and write the equation.

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$a|2-(-1)| + 5 = -4$ → $3a = \mathbf{-9}$ → $a = \mathbf{-3}$ $f(x) = \mathbf{-3|x + 1| + 5}$

A second one. Vertex (4, 1), passes through (7, 10). Find a and write the equation.

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$a|7-4| + 1 = 10$ → $3a = \mathbf{9}$ → $a = \mathbf{3}$ $f(x) = \mathbf{3|x - 4| + 1}$

Part 3 — Connecting Back to Day 2's Factoring

$f(x) = 2|3x - 12| + 5$ has a b-value hiding inside. Factor to find the vertex form $a|x - c| + d$.

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$2|3x - 12| + 5 = 2|3(x - 4)| + 5 = 2 \cdot \mathbf{3} \cdot |x - 4| + 5 = \mathbf{6}|x - 4| + 5$ → vertex **(4, 5)**

Part 4 — Real World: Dunhill Drafting's Roofline

Roofline data was modeled by trial-and-error transformation, landing on f(x) = -1.2|x - 3| + 8.

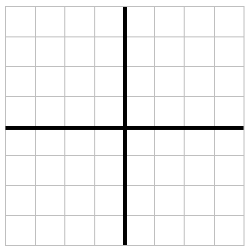
Vertex (highest point of the roof): **(3, 8)** Explain how you know it is a MAXIMUM, not a minimum, without graphing.

a = -1.2 < 0, so the graph opens downward and (3, 8) is the highest point

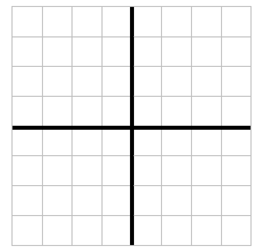
Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A. f(x) = |x - 2| - 3



B. f(x) = |x + 3| + 2



Practice

1. Give the vertex of f(x) = -3|x + 8| - 6. **(-8, -6)**

2. Factor f(x) = 3|2x + 6| - 1 into vertex form and give the vertex. **6|x + 3| - 1** vertex **(-3, -1)**

3. A function has vertex (-2, 8) and passes through (1, -1). Find a and write the equation.

   a|1-(-2)| + 8 = -1 → 3a = **-9** → a = **-3** f(x) = **-3|x + 2| + 8**

4. Factor f(x) = 5|10x - 30| + 2 into vertex form, and give both the vertex and the real a-value. **50|x - 3| + 2**
vertex **(3, 2)** a = **50**

Exit Ticket

A function has vertex (2, -5) and passes through (0, -1). Find a and write the equation, then explain how you know (2, -5) is a minimum.

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a|0-2| + (-5) = -1 → a = **2** f(x) = **2|x - 2| - 5** **a>0, opens up, so (2,-5) is the lowest point**